

Category Theory and Homological Algebra

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Categories

A **category** $\mathcal{C}$ consists of the following data:

- a collection $\text{Ob}(\mathcal{C})$ called the **objects** of $\mathcal{C}$,
- for each $A, B \in \text{Ob}(\mathcal{C})$, a collection $\text{Hom}_{\mathcal{C}}(A, B)$, called the **morphisms** from A to B ,
- a morphism $1_A \in \text{Hom}_{\mathcal{C}}(A, A)$ for each $A \in \text{Ob}(\mathcal{C})$
- for each $A, B, C \in \text{Ob}(\mathcal{C})$, a function called **composition**,

$$\circ : \text{Hom}_{\mathcal{C}}(A, B) \times \text{Hom}_{\mathcal{C}}(B, C) \longrightarrow \text{Hom}_{\mathcal{C}}(A, C)$$

$$f, g \longmapsto g \circ f.$$

with the axioms that composition is associative, and for each morphism $f \in \text{Hom}_{\mathcal{C}}(A, B)$, $\text{id}_B \circ f = f = f \circ \text{id}_A$.

Example. We give three examples of categories.

- “Set” has sets as objects and set functions as morphisms.
- Let R be a ring. “ R –Mod” has left R -modules as objects and R -module homomorphisms as morphisms.
- The category “Top” has topological spaces as objects and continuous maps as morphisms. “Haus” is the subcategory of “Top” consisting of Hausdorff spaces.

We give two more definitions—special types of morphism that generalize injective/surjective functions respectively.

- A **monomorphism** (mono) is a morphism $f : B \hookrightarrow C$ such that whenever morphisms $g, h : A \rightarrow B$ satisfy $f \circ g = f \circ h$, then $g = h$.
- An **epimorphism** (epi) is a morphism $f : A \twoheadrightarrow B$ such that whenever morphisms $g, h : B \rightarrow C$ satisfy $g \circ f = h \circ f$, then $g = h$.

Example. In the categories “Set” and “ R –Mod”, monos and epis are exactly injections and surjections. In “Haus”, monos are still injections, but epis need not be surjective—they only need to have dense image.

Universal Properties

All objects live inside a category $\mathcal{C}$.

- An **initial object** is an object I such that for any object C , there exists a unique morphism $f : I \rightarrow C$.
- A **terminal object** is an object T such that for any object C , there exists a unique morphism $g : C \rightarrow T$.
- A **zero object** is an object 0 which is both initial and terminal.
- A **product** of a family $\{A_{\lambda}\}_{\lambda \in \Lambda}$ is a pair $(A, \{\pi_{\lambda}\}_{\lambda \in \Lambda})$ of an object A and morphisms $\pi_{\lambda} : A \rightarrow A_{\lambda}$ (called *projection morphisms*) such that for any object T and morphisms $\{f_{\lambda} : T \rightarrow A_{\lambda}\}_{\lambda \in \Lambda}$, there exists a unique $f : T \rightarrow A$ such that Figure 1 commutes (for each $\lambda \in \Lambda$).
- A **coproduct** of a family $\{B_{\lambda}\}_{\lambda \in \Lambda}$ is a pair $(B, \{\iota_{\lambda}\}_{\lambda \in \Lambda})$ of an object B and morphisms $\iota_{\lambda} : B_{\lambda} \rightarrow B$ (called *inclusion morphisms*) such that for any object T and morphisms $\{g_{\lambda} : B_{\lambda} \rightarrow T\}_{\lambda \in \Lambda}$, there exists a unique $g : B \rightarrow T$ such that Figure 2 commutes (for each $\lambda \in \Lambda$).

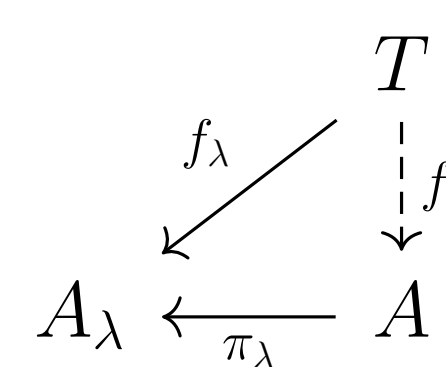


Fig. 1: Product

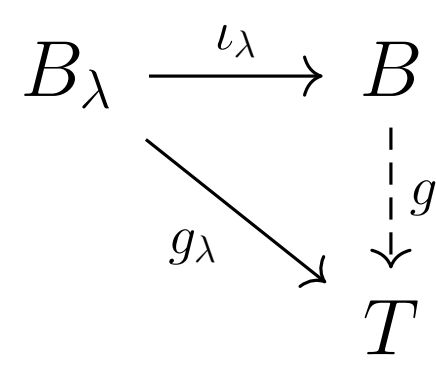


Fig. 2: Coproduct

Remark. In an arbitrary category objects satisfying these universal properties may not exist, but when they do exist, they are unique up to unique isomorphism.

Example. In “Set” (“Top” is similar),

- $\emptyset$ is initial.
- $\{*\}$ is terminal.
- There is no zero object.
- The product of $\{A_{\lambda}\}_{\lambda \in \Lambda}$ is the usual Cartesian product $\prod_{\lambda \in \Lambda} A_{\lambda}$ with componentwise projections.
- The coproduct of $\{B_{\lambda}\}_{\lambda \in \Lambda}$ is the *disjoint union* $\bigsqcup_{\lambda \in \Lambda} B_{\lambda}$ with inclusions.

Example. In “ R –Mod”,

- 0 is zero.
- The (finite) product and coproduct of $A_1, \dots, A_n$ is $A_1 \oplus \dots \oplus A_n$ with componentwise projections and inclusions.

Kernels and Cokernels

Let $\mathcal{C}$ be a category with zero object 0 . We say a morphism $g : A \rightarrow B$ is 0 if g is the composition of the unique morphisms $A \rightarrow 0$ and $0 \rightarrow B$.

- A **kernel** of a morphism $f : A \rightarrow B$ is a pair (K, k) of an object K and a morphism $k : K \rightarrow A$ such that $f \circ k = 0$ and for any object K' and morphism $k' : K' \rightarrow A$ such that $f \circ k' = 0$, there exists a unique morphism $u : K' \rightarrow K$ such that Figure 3 commutes.
- A **cokernel** of a morphism $f : B \rightarrow A$ is a pair (Q, q) of an object Q and a morphism $q : A \rightarrow Q$ such that $q \circ f = 0$ and for any object Q' and morphism $q' : A \rightarrow Q'$ such that $q' \circ f = 0$, there exists a unique morphism $u : Q \rightarrow Q'$ such that Figure 4 commutes.
- An **image** of a morphism $f : A \rightarrow B$ is a kernel of the cokernel of f .
- A **coimage** of a morphism $f : A \rightarrow B$ is a cokernel of the kernel of f .

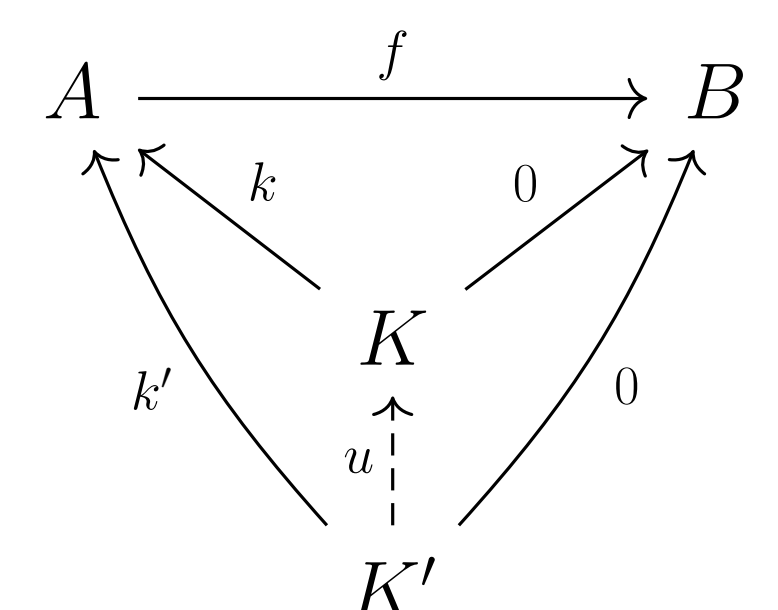


Fig. 3: Kernel

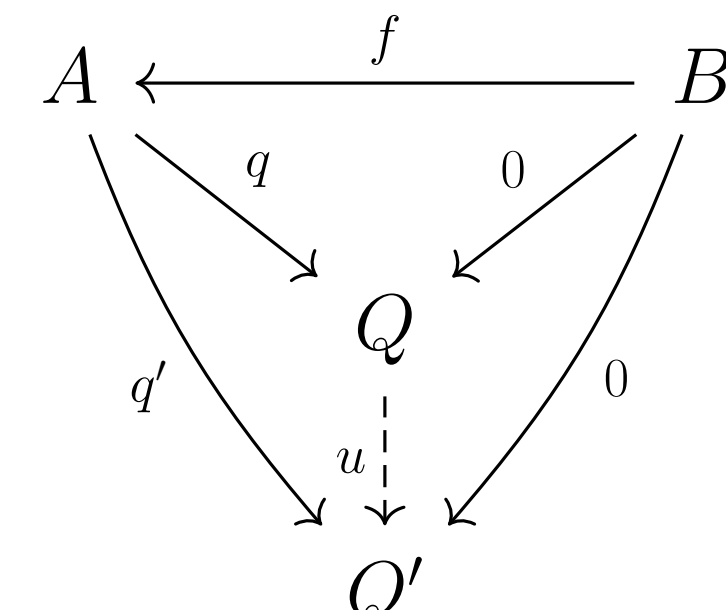


Fig. 4: Cokernel

Example. In the category R –Mod, a kernel of $f : A \rightarrow B$ is just $(\ker f, \iota)$, where ι is the inclusion of $\ker f \hookrightarrow A$. A cokernel of $f : A \rightarrow B$ is just $(B/\text{im } f, \pi)$, where π is the quotient map $B \rightarrow B/\text{im } f$.

Abelian Categories

- A **preadditive category** is a category $\mathcal{C}$ where all collections $\text{Hom}_{\mathcal{C}}(A, B)$ are endowed with an abelian group structure, and all morphisms $f, g, h \in \text{Hom}_{\mathcal{C}}(A, B)$ satisfy
 - * $f \circ (g + h) = (f \circ g) + (f \circ h)$
 - * $(f + g) \circ h = (f \circ h) + (g \circ h)$.
 i.e., $\circ$ is biadditive.
- An **additive category** is a preadditive category that has all finite products and a zero object. In an additive category, all finite products are also coproducts.
- An **abelian category** is an additive category where
 - * every morphism $f : A \rightarrow B$ has a kernel and cokernel.
 - * every monomorphism is a kernel, and every epimorphism is a cokernel.

- An **exact sequence** is a sequence

$$\dots \rightarrow X_{i-1} \xrightarrow{f_{i-1}} X_i \xrightarrow{f_i} X_{i+1} \rightarrow \dots$$

of objects and morphisms in an abelian category $\mathcal{A}$ such that $\text{im } f_i = \ker f_{i+1}$ for all i . In particular, $f_{i+1} \circ f_i = 0$.

Example. “ R –Mod” is an abelian category. “Set” and “Top” are not abelian (they do not even have a zero object).

Proposition. Let $f : A \rightarrow B$ in an abelian category $\mathcal{A}$. Then, f is a mono iff $\ker f = 0$, and f is an epi iff $\text{coker } f = 0$.

The Five Lemma

Theorem. (The Five Lemma) Let $\mathcal{A}$ be an abelian category. Consider the commutative diagram (Figure 5) of objects and morphisms in $\mathcal{A}$.

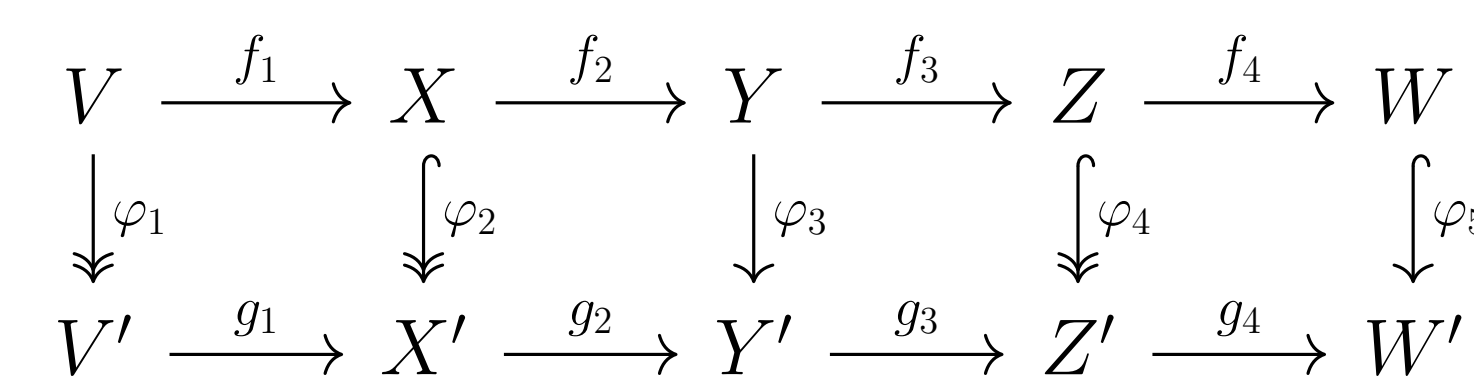


Fig. 5: The Five Lemma

Suppose both rows of the diagram are exact sequences.

Additionally, suppose that φ_2 and φ_4 are isomorphisms (both an epimorphism and a monomorphism), φ_1 is an epimorphism, and φ_5 is a monomorphism. Then φ_3 must be an isomorphism.

Proof. For simplicity, we shall assume $\mathcal{A}$ is the category of R -modules. To prove that φ_3 is an isomorphism, it suffices to show that φ_3 is both injective and surjective.

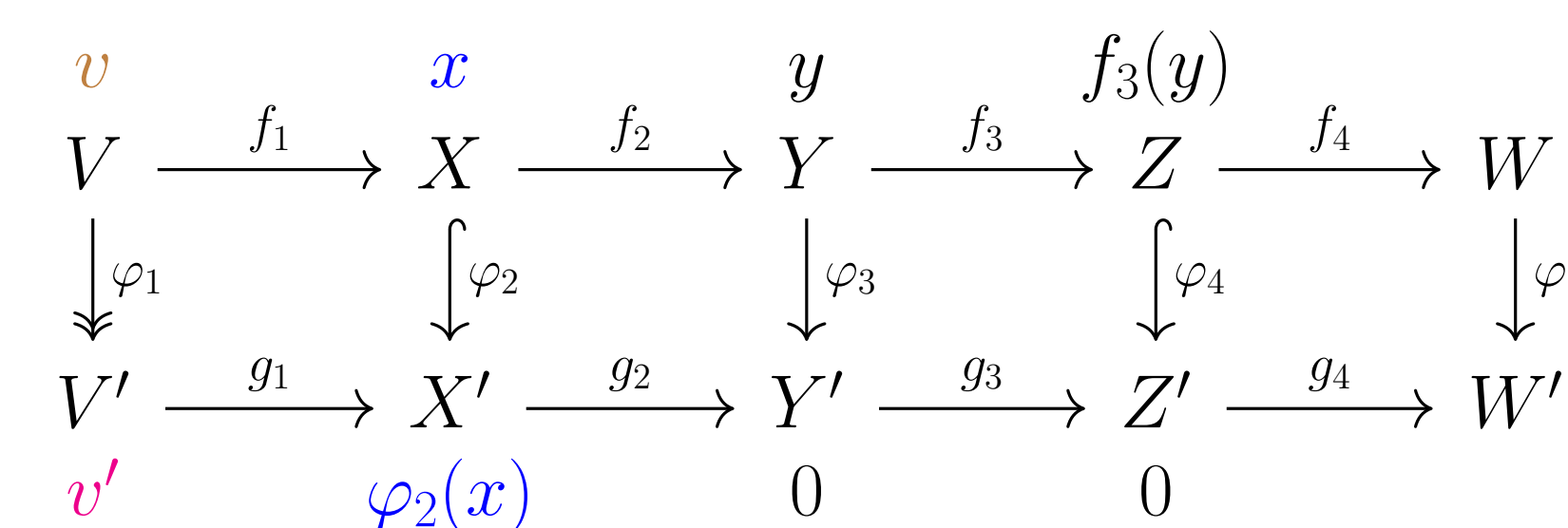


Fig. 6: Proof of Injectivity

To show injectivity of φ_3 , we prove that $\ker(\varphi_3) = \{0\}$. Let $y \in \ker(\varphi_3)$, so that $\varphi_3(y) = 0 \in Y'$. Then $g_3(\varphi_3(y)) = g_3(0) = 0$. Since the diagram commutes, $\varphi_4(f_3(y)) = g_3(\varphi_3(y)) = 0$. Since φ_4 is injective, $\ker(\varphi_4) = \{0\}$. Thus $f_3(y) = 0$, so $y \in \ker(f_3)$. Exactness of the first row tells us that $\text{im}(f_2) = \ker(f_3)$, so there exists an $x \in X$ such that $f_2(x) = y$. Now by commutativity of the diagram, we have

$$g_2(\varphi_2(x)) = \varphi_3(f_2(x)) = \varphi_3(y) = 0.$$

Thus $\varphi_2(x) \in \ker(g_2) = \text{im}(g_1)$, so there exists $v' \in V'$ such that $g_1(v') = \varphi_2(x)$. Surjectivity of φ_1 implies that there exists a $v \in V$ such that $\varphi_1(v) = v'$. Then,

$$\varphi_2(f_1(v)) = g_1(\varphi_1(v)) = g_1(v') = \varphi_2(x).$$

By injectivity of φ_2 , $f_1(v) = x$, so $f_2(f_1(v)) = f(x) = y$. By exactness, $f_2 \circ f_1$ is zero, so $y = 0$. Thus $\ker(\varphi_3) = \{0\}$, so φ_3 is injective.

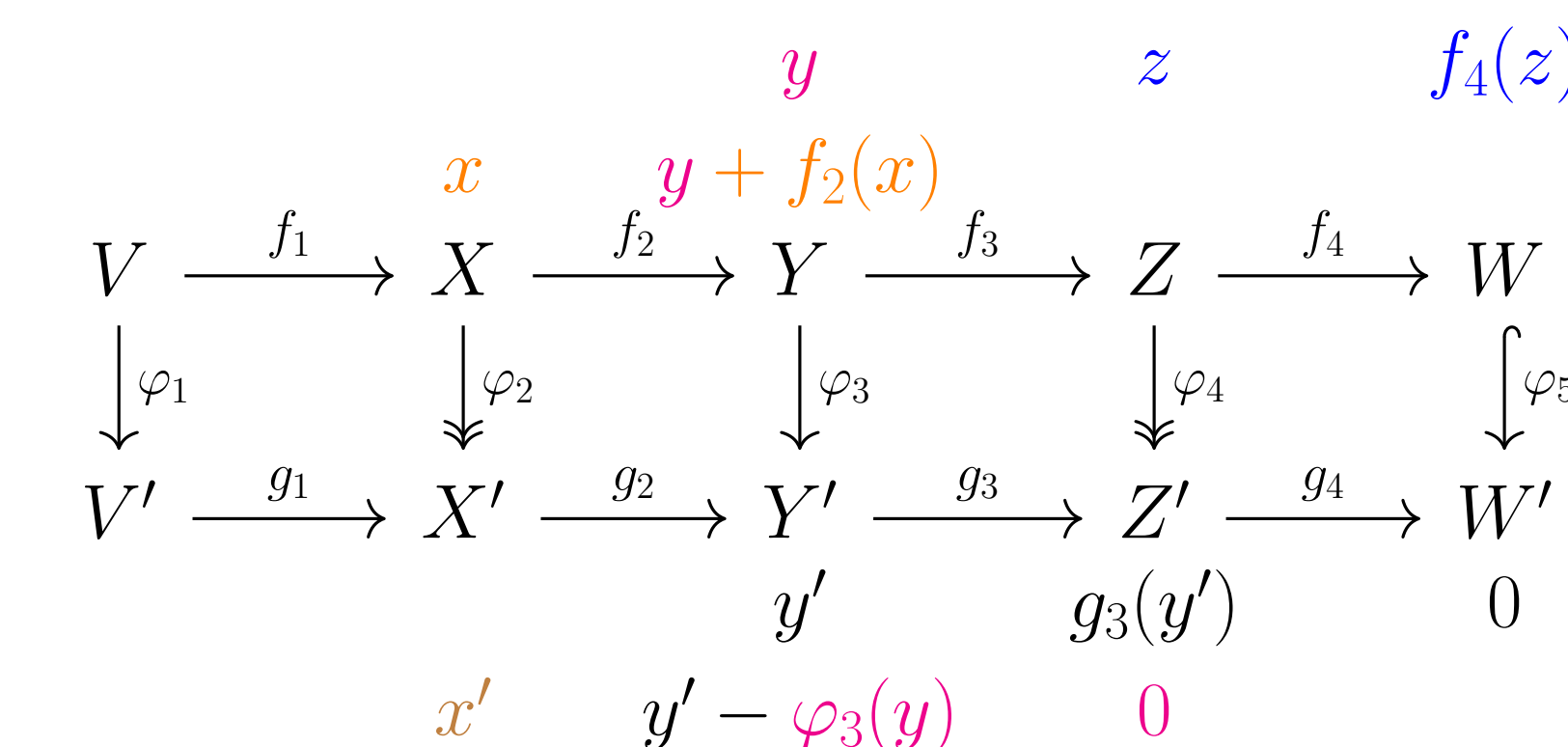


Fig. 7: Proof of Surjectivity

To prove φ_3 is surjective, let $y' \in Y'$. By surjectivity of φ_4 there exists $z \in Z$ such that $\varphi_4(z) = g_3(y')$. By exactness of the bottom row, $g_4 \circ g_3$ is zero. Then, by commutativity of the diagram,

$$0 = g_4(g_3(y')) = g_4(\varphi_4(z)) = \varphi_5(f_4(z)).$$

Thus $f_4(z) \in \ker(\varphi_5)$. Since φ_5 is injective, $f_4(z) = 0$. Thus, $z \in \ker(f_4) = \text{im}(f_3)$, so there exists $y \in Y$ such that $f_3(y) = z$. Then,

$$g_3(\varphi_3(y)) = \varphi_4(f_3(y)) = \varphi_4(z) = g_3(y').$$

Thus $g_3(y' - \varphi_3(y)) = 0$ because g_3 is a homomorphism. Then $y' - \varphi_3(y) \in \ker(g_3) = \text{im}(g_2)$ so there exists $x' \in X'$ such that $g_2(x') = y' - \varphi_3(y)$. Since φ_2 is surjective, there exists $x \in X$ such that $\varphi_2(x) = x'$. Then $\varphi_3(f_2(x)) = g_2(\varphi_2(x)) = g_2(x') = y' - \varphi_3(y)$. Therefore, $\varphi_3(f_2(x) + y) = \varphi_3(f_2(x)) + \varphi_3(y) = y' - \varphi_3(y) + \varphi_3(y) = y'$. Thus, every element of Y' has a nonempty preimage in Y , and therefore φ_3 is surjective. $\square$

References

- (1) Francis Borceux. The Handbook of Categorical Algebra I: Basic Category Theory. Cambridge University Press, 1994.
- (2) Thomas W. Hungerford. Algebra. Springer Science & Business Media, 2012.
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